

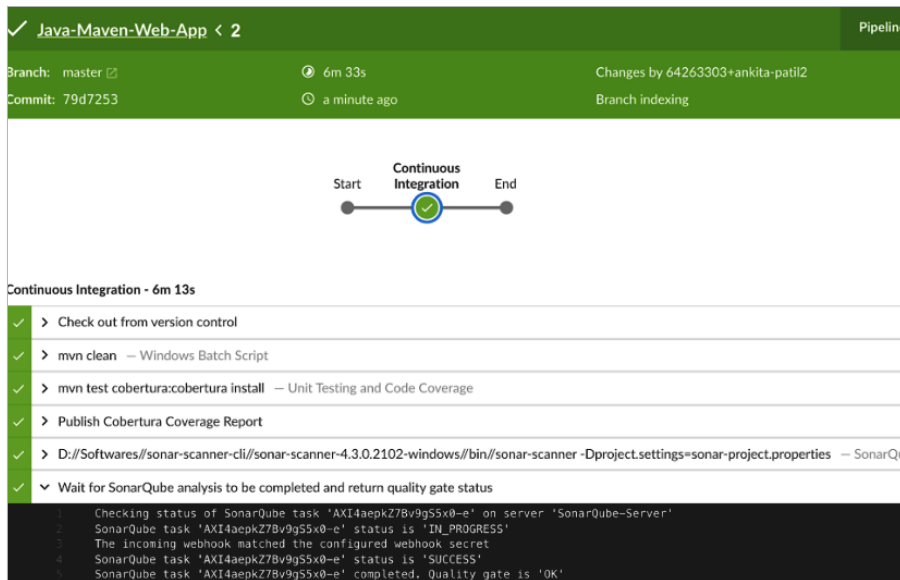


Figure 5: SonarQube 'quality gate' check in Jenkins pipeline

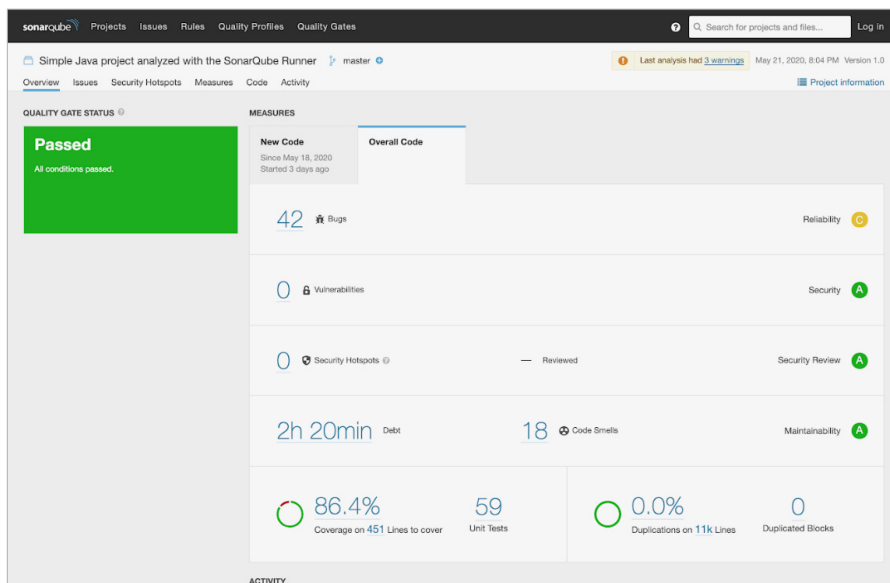


Figure 6: SonarQube project dashboard after analysis from Jenkins

1. **Reliability issues:** Bugs — these are coding errors that will break your code at some point of time and need to be fixed immediately.
2. **Security issues:** Vulnerabilities and security hotspots are issues in your code that are open to attack.
3. **Maintainability issues:** Code smells make your code confusing and difficult to maintain. If these issues are not taken care of, they can lead to high technical debt.

Whenever you see issues in the SonarQube project, you need to click on the *Issues* tab besides *Overview*, to view and navigate through them and look at what is causing them. This will help you to change state or

priority, comment, or assign the issue to the team.

On the dashboard, we also see code coverage and unit test cases. These are the result of the code coverage report generated by Jacoco and Cobertura and of the unit test results generated after successful execution of the JUnit test case. If you click on *Code coverage percent* it will take you to code that is covered by unit test cases.

SonarQube also analyses duplication of code in the files being looked at. This is also published to the dashboard.

In the first part of this two-part series, we learnt that static code analysis is an important part of the software development life cycle (SDLC) to build the best quality software. It has lots of benefits if it is adopted from the start of the software development process. We also covered the top five tools available to perform static code analysis on your code.

SonarQube is one of the most widely used tools for code analysis. In this article, we have covered how to configure SonarLint, GitHub app, and pull request decoration with GitHub and SonarQube. Lastly, we created a Jenkins pipeline to perform SonarQube analysis on a Maven application. **END** 🐧

References

- [1] SonarQube Documentation, <https://docs.sonarqube.org/latest/>
- [2] Hands-on Pipeline as Code with Jenkins, <https://www.amazon.com/Hands-Pipeline-Code-Jenkins-Implementation/dp/9389898609>

By: Ankita Patil

The author has written the book 'Hands-on Pipeline as Code with Jenkins: CI/CD Implementation for Mobile, Web and Hybrid Applications Using Declarative Pipeline in Jenkins'.